SAFETY MEASURES AND FMEA

1. PHYSICAL FAIL SAFES:-

ELECTRICAL FAIL SAFES:-

• **Voltage divider ahead of GPIO34 (2x 100kW):** halves the battery voltage before it reaches the ESP32's analog pin, keeping it inside the chip's 3.3V absolute maximum input rating at every charge state.

• **Physical ON/OFF slide switch (SPST):** lets the device be fully powered down without disconnecting the battery lead, so it is not silently drawing current while not being worn.

• **Protected (not bare) TP4056 module specified in the BOM:** deliberately chosen over the cheaper unprotected variant specifically because the unprotected version has no overcharge or overdischarge protection at all.

MECHANICAL FAIL SAFES:-

• **Rounded enclosure corners (3mm radius):** removes sharp edges on a device worn against skin for an extended period.

• **Skin-contact liner (TPU/silicone):** separates the rigid PETG enclosure and electronics from direct skin contact.

• **Battery placed away from the skin-contact window:** positioned under the main board on the outward side, reducing the chance of localised heat or pressure against skin if the cell were ever to warm up.

• **Adjustable strap (22–34cm range):** avoids a fixed-size band being worn too tight, which could restrict circulation overnight

DATA, FIRMWARE FAIL SAFES:-

(planned behaviour, to be implemented in firmware)

• Motion/wear data from the MPU6050 is used to flag or discard nights where the band was not worn correctly, rather than silently logging unreliable readings as if they were valid.

• A data-quality score per night (based on wear-contact consistency and motion level) is intended to gate whether a night's data is used in analysis, so a bad night does not quietly corrupt a cycle estimate

2. FAILURE MODE AND EFFECT ANALYSIS (FMEA):-

Failure mode	Potential effect / hazard	Mitigation already in design
LiPo cell physically punctured or crushed	Internal short, thermal event	Enclosure fully encloses the cell in a dedicated pocket away from the skin-contact window; strap tension does not bear on the battery pocket
Band loses skin contact during the night (slips, strap loosens)	Inaccurate or missing temperature reading for that night	MPU6050 motion/wear data used to flag low-quality nights; night excluded from cycle-phase analysis rather than treated as valid
ESP32 firmware hangs or crashes mid-night	Loss of that night's logged data	Battery protection circuit is hardware-level and independent of firmware state, so a frozen MCU cannot cause a battery safety failure;

		data loss itself is a functional risk, addressed in Section 3
I2C bus conflict (two sensors sharing the same address)	One or more sensors unreadable, silent data gaps	All four sensor addresses individually verified against datasheets (0x48, 0x38, 0x57, 0x68) and confirmed non-overlapping before wiring
MAX30102 pull-ups wired to internal 1.8V rail instead of 3.3V (known issue on some modules)	Sensor not detected on the 3.3V I2C bus, no heart-rate/respiration data	Flagged explicitly in the BOM's must-contain check; to be confirmed by I2C scan on arrival, with external 4.7k pull-ups to 3.3V as a documented fallback if the onboard ones are wrong
Battery pack lacks its own built-in protection circuit	Reduced redundancy if the TP4056 protection stage were ever bypassed or miswired	BOM explicitly requires the battery listing to state PCM protection before purchase
Charging while still being worn	Mild heating near the charging circuit against skin during charge	Device is charged only while off the arm, consistent with the seller guidance noted in the BOM to disconnect the load while charging; slide switch used to power down before charging

3. TECHNICAL BOTTLENECKS AND BACKUP SOLUTIONS:-

Bottleneck	Why it's a risk	Backup solution
MAX30205 default I2C address uncertain before testing (0x48 vs 0x49)	An early design draft assumed 0x49; the datasheet's address-pin scheme gives 0x48 with all address pins low	Resolved by running an I2C bus scan once the physical module arrives, rather than assuming a value; firmware reads whichever address the scan reports
Several module listings do not state onboard pull-up resistor values or decoupling capacitors	A module could ship without adequate pull-ups, causing unreliable I2C communication	Carrier board independently includes its own 4.7k SDA/SCL pull-ups and 0.1µF decoupling capacitors per sensor, regardless of what each module already has, so the shared bus works even if a module's onboard components are inadequate or absent
ESP32 listing sourced is the 38-pin layout, not the 30-pin DevKit V1	A different pin layout could misalign the schematic's	Confirmed that the pins actually used by this design (3V3, GND, EN, GPIO21, GPIO22, GPIO0, GPIO2, GPIO34) exist

originally assumed	header connections	on both layouts; header footprint in the carrier board to be matched to the 38-pin variant specifically once confirmed on arrival
Skin-contact temperature accuracy depends on consistent sensor-to-skin contact overnight	Movement or a loose strap could introduce noise into the one signal the whole device depends on	Foam block in the CAD design presses the sensor against skin with gentle constant pressure; MPU6050 flags low-contact nights for exclusion rather than silently accepting bad data